

Raghav Kochhar

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Summary

AI/ML Engineer building streaming ML systems on GCP. Built Pub/Sub/Dataflow pipelines and Cloud Run services, reducing streaming costs for Waste-to-Energy.

Education

Vivekananda Institute of Professional Studies – Technical Campus 2021 – 2025
Bachelor of Technology in Artificial Intelligence and Machine Learning (CGPA: 9.03/10).

Experience

Artificial Intelligence and Machine Learning Engineer, Jaipur Robotics August 2025 – Present

- Streaming and Data: Built a Dataflow pipeline (Apache Beam, Python) that merges sensor streams across cameras and handles duplicate detections; reduced costs with right-sizing and built reliable AlloyDB updates for truck sessions.
- Computer Vision: Built services for image segmentation and alert image processing; contributed to LiDAR volume measurement.
- GenAI and Automation: Built alert visibility tracking and contributed to LLM classification, truck video analysis, material mixing analysis, and automated report generation.
- Platform: Built a video-generation platform (FastAPI + Streamlit, GCS, Cloud Tasks) for timelapse videos and worked with Django data models (AlloyDB) for truck tracking.

Projects

American Sign Language Interpreter

- Built a real-time sign language recognition system with EfficientNet and LSTM.
- Deployed the model via a FastAPI backend and Streamlit interface with live webcam inference.

Speech Emotion Classifier

- Developed a seven-class emotion recognition LSTM model trained on audio samples.
- Built a FastAPI inference API and Streamlit UI for real-time and batch audio analysis.

Traffic & Pedestrian Analyzer

- Created a YOLOv8-powered traffic monitoring system with pedestrian and signal detection.
- Integrated traffic-light color recognition for smart-city insights.

Automaton: No-Code ML Workflow Platform

- Implemented a No-Code ML workflow platform for data preprocessing, training, and tuning.
- Built interactive dashboards for model benchmarking and performance visualization.

Research (Under Review, PDF available upon request)

Publication: Efficient Adaptation of Lightweight LLMs (ICAAI 2025, Springer)

- Benchmarked Baseline, T-Free, and STELLA on consumer-grade hardware, validating edge feasibility.
- Achieved 75% embedding reduction with T-Free while maintaining accuracy.

Publication: Health Helix: Connecting You to Better Health (COM-IT-CON 2024)

- Presented at the International Conference on Progressive Computational Intelligence (Taylor & Francis Group); architected a unified healthcare platform.
- Designed engagement workflows that reduced clinic no-show rates via automated AI alerts and scheduling integration.

Technical Skills

- Languages: Python (uv, ruff, mypy, pytest), SQL, Bash.
- ML and Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, ONNX, Scikit-learn.
- LLMs and GenAI: LangChain, RAG pipelines, Prompt Engineering, Vertex AI.
- Computer Vision and NLP: OpenCV, YOLOv8, FFmpeg.
- Web and APIs: FastAPI, Streamlit, SQLAlchemy, Pydantic.
- Data Engineering: Apache Beam, BigQuery, PySpark, Pandas, NumPy.
- Cloud and DevOps: GCP (Vertex AI, Pub/Sub, Dataflow, Cloud Run, GKE), Docker, GitHub Actions, Cloud Build.